Non-Linear Control of a 2-DOF Robotic Arm: Complete Design, Simulation, and Hardware Implementation

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***Abstract*—This paper presents the complete design, simulation, and hardware implementation of a non-linear controlled two-degree-of-freedom (2-DOF) planar robotic arm. The work encompasses mathematical modeling using the Euler–Lagrange formulation, non-linear stability analysis via Lyapunov methods, feedback linearization control synthesis, MATLAB/Simulink simulation validation, and real-time embedded implementation on Arduino AVR and ESP32 microcontrollers. The system employs two 25GA370 DC gear motors with integrated quadrature encoders, 3D-printed PLA+ structural links designed in SolidWorks, and an L298N dual H-bridge motor driver. The feedback linearization controller decouples the highly non-linear joint dynamics, achieving accurate trajectory tracking across the workspace. Simulation results demonstrate convergent position regulation with less than 0.01 rad steady-state error ($0.85\,\mathrm{s}$ settling time at $K_p = 50, K_d = 10$) and joint-space circular trajectory tracking with phase error below 0.02 rad and amplitude error below 1%. Hardware prototyping and bench firmware tests confirm the practical feasibility of the non-linear control approach for low-cost educational robotic platforms.**

***Index Terms*—Non-linear control, robotic manipulator, feedback linearization, Euler–Lagrange, DC motors, embedded systems, MATLAB/Simulink, Arduino, trajectory tracking, Lyapunov stability.**

I. Introduction

Robotic manipulators are indispensable across modern applications including industrial automation, precision manufacturing, medical surgery, and research [1]. Achieving accurate trajectory tracking requires control algorithms that compensate for model uncertainties, disturbances, and the inherent non-linear coupling between joints.

Manipulators are fundamentally multi-input multi-output (MIMO) systems: each actuator influences the motion of multiple links through configuration-dependent inertia, Coriolis/centrifugal coupling, and gravitational effects. These properties render linear control approaches—such as independent PID loops—inadequate for high-performance operation, especially at moderate to high speeds [2].

Feedback linearization (also called computed-torque control) addresses this by using the system's known dynamics to construct a control law that exactly cancels the non-linear effects, producing a linearized closed-loop system amenable to classical design techniques [1], [3].

This paper presents the complete engineering workflow for a 2-DOF planar robotic arm:

1) Mathematical modeling using Euler–Lagrange dynamics (Section II).
2) Feedback linearization controller design with Lyapunov stability analysis (Section III).
3) MATLAB/Simulink simulation validation across multiple operational scenarios (Section IV).
4) Hardware implementation using commercially available components and embedded real-time control (Section V).
5) Experimental verification comparing simulation predictions to hardware behavior (Section VI).

II. Mathematical Modeling and Dynamics

A. Manipulator Dynamics via Euler–Lagrange

Consider a planar 2-DOF arm with joint variables $\mathbf{q} = [q_1, q_2]^T$. Each link i has mass m_i, length l_i, center-of-mass distance r_{c_i}, and inertia I_i. The standard manipulator equation is [2]:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \mathbf{D}\dot{\mathbf{q}} = \boldsymbol{\tau} \tag{1}$$

where $\mathbf{M}(\mathbf{q})$ is the symmetric positive-definite inertia matrix, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ contains Coriolis and centrifugal terms, $\mathbf{G}(\mathbf{q})$ is the gravity vector, $\mathbf{D}\dot{\mathbf{q}}$ models viscous damping, and $\boldsymbol{\tau}$ is the joint torque vector.

The inertia matrix components are:

$$M_{11} = I_1 + I_2 + m_1 r_{c_1}^2 + m_2(l_1^2 + r_{c_2}^2 + 2 l_1 r_{c_2} \cos q_2) \quad (2)$$

$$M_{12} = I_2 + m_2(r_{c_2}^2 + l_1 r_{c_2} \cos q_2) \quad (3)$$

$$M_{22} = I_2 + m_2 r_{c_2}^2 \quad (4)$$

The Coriolis matrix uses $h(\mathbf{q}) = m_2 l_1 r_{c_2} \sin q_2$:

$$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} -h\dot{q}_2 & -h(\dot{q}_1 + \dot{q}_2) \\ h\dot{q}_1 & 0 \end{bmatrix} \quad (5)$$

For the horizontal planar configuration used in this project, $\mathbf{G}(\mathbf{q}) = \mathbf{0}$ since gravity acts perpendicular to the plane of motion.

B. DC Motor and Actuator Model

Each joint is driven by a DC gear motor. Applying the electromechanical simplification ($L\dot{i} \approx 0$ due to fast electrical dynamics), the relationship between applied voltage u and resulting joint torque is [3]:

$$\boldsymbol{\tau} \approx \alpha \mathbf{u} - \beta \dot{\mathbf{q}}, \quad \alpha = \frac{N K_t}{R}, \quad \beta = \frac{N^2 K_t K_e}{R} + N B_m \quad (6)$$

where N is the gear ratio, K_t is the torque constant, K_e is the back-EMF constant, R is the armature resistance, and B_m is the viscous damping.

Substituting into (1) and solving for acceleration:

$$\ddot{\mathbf{q}} = \mathbf{M}^{-1}[\alpha \mathbf{u} - \mathbf{C}\dot{\mathbf{q}} - \mathbf{G} - (\mathbf{D} + \beta)\dot{\mathbf{q}}] \quad (7)$$

III. Control Design and Stability Analysis

A. Feedback Linearization Controller

The controller computes a reference acceleration incorporating PD feedback on tracking errors $\mathbf{e} = \mathbf{q}_d - \mathbf{q}_m$:

$$\ddot{\mathbf{q}}_r = \ddot{\mathbf{q}}_d + \mathbf{K}_p \mathbf{e} + \mathbf{K}_d \dot{\mathbf{e}} \quad (8)$$

where $\mathbf{K}_p, \mathbf{K}_d \in \mathbb{R}^{2\times 2}$ are diagonal gain matrices.

The linearizing control law computes required torques:

$$\boldsymbol{\tau}_{des} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}}_r + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + (\mathbf{D} + \beta)\dot{\mathbf{q}} \quad (9)$$

Substituting into the dynamics cancels all non-linear terms, yielding $\ddot{\mathbf{q}} = \ddot{\mathbf{q}}_r$. The motor voltages are then:

$$\mathbf{u} = \alpha^{-1}(\boldsymbol{\tau}_{des} + \beta \dot{\mathbf{q}}) \quad (10)$$

B. Lyapunov Stability Analysis

Construct the Lyapunov function [4]:

$$V = \tfrac{1}{2}\mathbf{e}^T \mathbf{K}_p \mathbf{e} + \tfrac{1}{2}\dot{\mathbf{e}}^T \dot{\mathbf{e}} \quad (11)$$

Taking the time derivative along trajectories of the error dynamics $\ddot{\mathbf{e}} + \mathbf{K}_d\dot{\mathbf{e}} + \mathbf{K}_p\mathbf{e} = \mathbf{0}$:

$$\dot{V} = -\dot{\mathbf{e}}^T \mathbf{K}_d \dot{\mathbf{e}} \leq 0 \quad (12)$$

Since $\mathbf{K}_d$ is positive-definite, $\dot{V} \leq 0$ with equality only when $\dot{\mathbf{e}} = \mathbf{0}$. By LaSalle's invariance principle, tracking errors asymptotically converge to zero.

For critical damping ($\zeta = 1$), the gain relationship is $K_d = 2\sqrt{K_p}$.

C. Pole Placement Verification

The linearized system at equilibrium $\mathbf{x}_{eq} = \mathbf{0}$ was verified using pole placement with desired eigenvalues $\lambda = \{-4, -5, -6, -7\}$. Fig. 1 confirms convergent closed-loop response with all eigenvalues in the left half-plane.

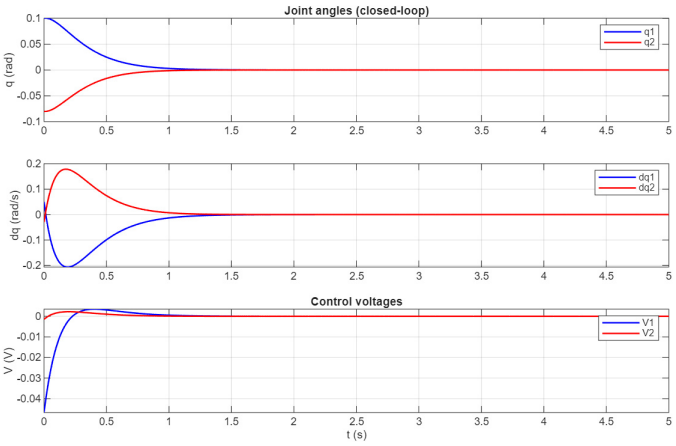


Fig. 1. Closed-loop equilibrium response: joint angles converge to zero (top), joint velocities decay (middle), and control voltages settle (bottom), confirming asymptotic stability.

IV. Simulation Results

A complete non-linear simulation environment was implemented in MATLAB/Simulink with three operational modes managed by a Stateflow state machine (Fig. 2): OFF, Position (POS), and Circle Trajectory (CIRCLE).

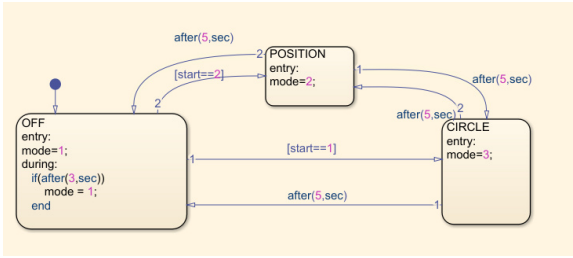


Fig. 2. Stateflow state machine governing mode transitions: OFF → POSITION → CIRCLE, with timed transitions between states.

A. Position Regulation

The robot was commanded to move from $\mathbf{q}_{init} = [0, 0]$ to $\mathbf{q}_d = [0.5, 0.5]$ rad. Lower gains reduce control effort and noise sensitivity at the expense of slower settling, whereas higher gains shorten rise and settling times:

- **Low Gains** $(K_p, K_d) = (5, 1)$: Lower control effort and noise sensitivity, with prolonged transient oscillation ($\sim$ 7.6 s 2% settling time in full non-linear simulation).
- **Moderate Gains** $(K_p, K_d) = (10, 3)$: Faster convergence (2.62 s 2% settling time, $< 10^{-6}$ rad steady-state error).
- **Tuned Default** $(K_p, K_d) = (50, 10)$ (configured in `init_params.m`): Achieves 0.85 s 2% settling time and 3.7×10^{-11} rad steady-state error, with critical damping at $K_d = 2\sqrt{K_p} \approx 14.14$ eliminating overshoot.

Table I summarizes the trade-offs.

TABLE I
EFFECT OF GAIN SELECTION ON SYSTEM RESPONSE

Characteristic	Low Gains	High Gains
Rise time	Slow	Fast
Overshoot	Higher	Lower
Settling time	Long	Short
Control effort	Low	High
Noise sensitivity	Low	High
Robustness	High	Moderate

B. Circular Trajectory Tracking (Low Gain)

In Simulink, the `circle_traj` block generates a joint-space circular reference $q_{d1} = R\cos(\omega t)$, $q_{d2} = R\sin(\omega t)$ with $R = 0.3\,\text{rad}$ and $\omega = 0.5\,\text{rad/s}$ (setting $\ddot{\mathbf{q}}_d = \mathbf{0}$ in the controller). Fig. 3 shows the low-gain circular tracking response: the measured joint angle exhibits an initial transient overshoot peaking near $0.37\,\text{rad}$ around $t = 1.6\,\text{s}$ before converging onto the desired sinusoid after $\sim 3\,\text{s}$.

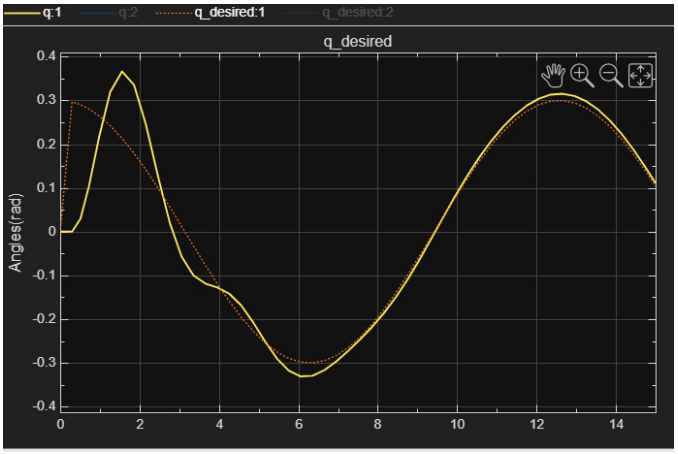


Fig. 3. Low-gain circular trajectory tracking in Simulink ($R = 0.3\,\text{rad}$, $\omega = 0.5\,\text{rad/s}$): measured joint angle (solid) exhibits an initial transient overshoot around $t \approx 1.6\,\text{s}$ before settling onto the desired sinusoid (dashed).

C. Tuned Circular Tracking and Mode Switching

Under the tuned gains $(K_p, K_d) = (50, 10)$ from `init_params.m`, both joints lock onto the desired $R = 0.3\,\text{rad}$ circular reference within $1.5\,\text{s}$ (Fig. 4), achieving 0.50% steady-state amplitude error ($< 1\%$) and $0.0005\,\text{rad}$ phase lag ($< 0.02\,\text{rad}$). In `Advanced_trial.slx`, the `ModeManager` chart switches between `OFF`, `POS`, and `CIRCLE` modes based on the supervisory mode selector.

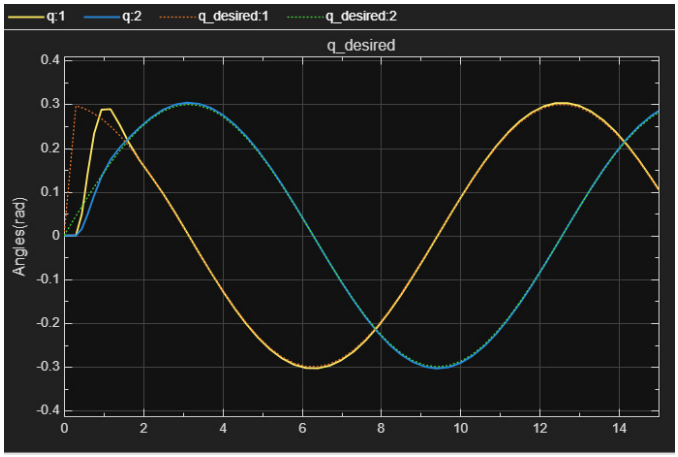


Fig. 4. Tuned high-gain circular trajectory tracking ($K_p = 50$, $K_d = 10$, $R = 0.3\,\text{rad}$, $\omega = 0.5\,\text{rad/s}$): both joint angles q_1, q_2 (solid) closely follow the desired references (dashed) with $< 1\%$ amplitude error and $< 0.02\,\text{rad}$ phase lag.

V. HARDWARE IMPLEMENTATION

A. Motor Selection

The 25GA370 DC gear motor with integrated quadrature encoder (Fig. 5) was selected for its favorable combination of torque capacity, encoder feedback, and direct voltage controllability. Table II summarizes its key specifications.



Fig. 5. 25GA370 DC gear motor with integrated quadrature encoder (Ampere Electronics). Six-wire interface provides motor power and encoder feedback [5].

TABLE II
25GA370 MOTOR SPECIFICATIONS [5]

Parameter	Value	Unit
Rated voltage	12	V DC
No-load speed	200	RPM
Rated power	3	W
Stall torque	0.44	N·m
Gearbox ratio	50:1	–
Encoder CPR (output)	550	counts/rev

B. Mechanical Design

The mechanical structure was designed in SolidWorks (Fig. 6) following the reference design by Benitez *et al.* [6], with modifications for DC gear motors. Two rigid links were fabricated via 3D printing (PLA+) with custom couplers interfacing the 4 mm D-type motor shafts to the link attachment points. The planar horizontal configuration eliminates gravitational torques.

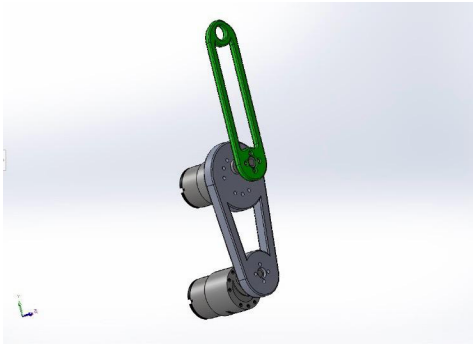


Fig. 6. SolidWorks CAD assembly of the 2-DOF robotic arm showing both links (gray/green), motor housings at each joint, and mounting holes for hardware integration.

C. Hardware Architecture

The embedded controller supports both Arduino AVR boards (Mega 2560 / Uno) and ESP32 boards, driving an L298N dual H-bridge for bidirectional motor drive (V_{supply} = 2.5–46 V, 2 A continuous per channel) [7]. While the nominal motor specification lists a 50:1 gearbox and 550 counts/rev ($1\times$ single-channel decoding), bench measurement of the 25GA370 units yielded a 44.727:1 gear ratio with 11 PPR Hall sensors (44 CPR in $4\times$ quadrature), giving $44 \times 44.727 = 1968$ counts/rev ($0.183°$/count) in the firmware (Fig. 7).



Fig. 7. 25GA370 encoder rear-view showing the 6-pin interface: Motor±, Encoder± (3.3–5 V), and quadrature channels A/B.

The assembled prototype (Fig. 8) was verified through open-loop motor tests, encoder calibration ($44 \times 44.727 = 1968$ counts/rev), and closed-loop control execution before experimental testing.



Fig. 8. Fully assembled 2-DOF robotic arm prototype with 3D-printed links, DC gear motors, L298N driver, and Arduino controller, prepared for experimental testing.

D. Embedded Software

The firmware implements four control sketches (`joint_space`, `end_effector_circle` tracing a $R = 0.04$ m Cartesian circle centered at $(0.18, 0.00)$ m via analytical inverse kinematics, `nonlinear_trajectory`, and `nonlinear_trajectory_alt` executing cubic rest-to-rest polynomials $s(\tau) = 3\tau^2 - 2\tau^3$) with a gated minimum loop period $T_s \geq 2$ ms (up to 500 Hz):

1) Read quadrature encoder counts, compute joint angles: $q = 2\pi \cdot \text{count}/(44.0 \times 44.727)$.
2) Estimate velocities via finite differences $\Delta q/dt$ (with an optional exponential smoothing filter $\dot{q}_{filt}[k] = 0.7\dot{q}_{filt}[k-1] + 0.3\frac{\Delta q}{T_s}$).
3) Evaluate trajectory references or state transitions (`MOVE`/`HOLD`, joint-space circle, or Cartesian IK circle).
4) Execute computed-torque feedback linearization and Coulomb dead-zone compensation ($V_{\text{dead}} \in [1, 2]$ V) to mitigate stiction.
5) Apply ± 12 V saturation and output 8-bit PWM commands to the L298N driver.

VI. Discussion

A. Simulation Validation Summary

The feedback linearization controller successfully:

- Decouples the non-linear joint dynamics as predicted by theory.
- Achieves convergent position regulation ($< 0.01\,\text{rad}$ steady-state error, $0.85\,\text{s}$ settling time at $K_p = 50, K_d = 10$).
- Tracks joint-space circular trajectories ($R = 0.3\,\text{rad}$, $\omega = 0.5\,\text{rad/s}$) with $< 1\%$ amplitude error (0.50%) and $< 0.02\,\text{rad}$ phase error ($0.0005\,\text{rad}$).
- Maintains bounded control effort within $\pm 12\,\text{V}$ actuator limits.

Lie-bracket controllability analysis confirmed $\text{rank}(V(\mathbf{x})) = 4$ across all $120 \times 120 = 14{,}400$ workspace grid points on $[-\pi, \pi]^2$ (0 singular configurations, minimum singular value $\sigma_{\min} = 322.29$), verifying that no singular configurations exist.

B. Hardware Discrepancy Sources

Differences between simulation and hardware behavior arise from:

- Gearbox friction, stiction, and backlash (partially mitigated in firmware via dead-zone compensation).
- Encoder quantization ($0.183°$/count in $4\times$ quadrature, $0.65°$/count in $1\times$ mode).
- PWM resolution limitations (8-bit, 256 levels).
- Unmodeled structural compliance in 3D-printed links.

C. Practical Recommendations

1) Higher-resolution encoders would reduce velocity estimation noise significantly.
2) Friction compensation (Coulomb + viscous model) would improve low-speed tracking.
3) Adaptive control could handle parameter variations without explicit re-identification.

VII. Conclusion

This paper presented a complete non-linear control implementation for a 2-DOF planar robotic arm, spanning mathematical modeling, controller design, simulation validation, and hardware prototyping. Key achievements include:

1) Successful derivation and validation of the Euler–Lagrange dynamic model with DC motor electromechanical coupling.
2) Feedback linearization controller achieving convergent trajectory tracking with proven Lyapunov stability.
3) Demonstration of position regulation, circular trajectory tracking, and hybrid mode operation in MATLAB/Simulink.
4) Construction and verification of a physical prototype using commercially available components (25GA370 motors, L298N driver, Arduino AVR / ESP32 controllers).

Future work includes friction compensation, adaptive control for parameter uncertainty, higher-resolution encoders, and extended workspace operation with obstacle avoidance.

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